Complex Number

308. Let z be a complex number such that $|z| = 1$. If $\frac{2+k^2z}{k+\bar{z}} = kz, k \in \mathbf{R}$ then the maximum distance of $k + ik^2$ from the circle $|z - (1 + 2i)| = 1$ is :

माना z एक सम्मिश्र संख्या है जिसके लिए $|z| = 1$ है। यदि $\frac{2+k^2z}{k+\bar{z}} = kz, k \in \mathbf{R}$ है, तो वृत्त $|z - (1 + 2i)| = 1$ से $k + ik^2$ की अधिकतम दूरी है :

(A) 3 (B) $\sqrt{3}+1$ (C) $\sqrt{5}+1$ (D) 2

Ans. C

Sol. $\frac{2+k^2z}{k+\bar{z}} = kz$

$|z|^2 k = 2$

$k = 2$

point p(2, 4); center (1, 2)

distance from circle

$(x - 1)^2 + (y - 2)^2 = 1$ is max.

if $(OP + r) = \sqrt{1+4}+1 = \sqrt{5}+1$

309. Let $z \in C$ be such that $\frac{z^2+3i}{z-2+i} = 2 + 3i$. Then the sum of all possible values of z^2 is :

माना $z \in C$ के लिए $\frac{z^2+3i}{z-2+i} = 2 + 3i$ है। तो z^2 के सभी संभव मानों का योग है :

(A) $-19 + 2i$ (B) $-19 - 2i$ (C) $19 - 2i$ (D) $19 + 2i$

Ans. B

Sol. $z^2 + 3i = z(2 + 3i) - 7 - 4i$

$z^2 - z(2 + 3i) + 7 + 7i = 0 \begin{matrix} \nearrow z_1 \\ \searrow z_2 \end{matrix}$

$z_1^2 + z_2^2 = (z_1 + z_2)^2 - 2z_1 z_2$

$= -19 - 2i$

310. If $z_1, z_2, z_3 \in C$ are the vertices of an equilateral triangle, whose centroid is z_0, then $\sum_{k=1}^{3}(z_k - z_0)^2$ is equal to :

माना एक समबाहु त्रिभुज के शीर्ष $z_1, z_2, z_3 \in C$ हैं। यदि इसका केन्द्रक z_0 है, तो $\sum_{k=1}^{3}(z_k - z_0)^2$ बराबर है :

(A) $-i$ (B) 1 (C) 0 (D) i

Ans. C

Sol. $z_1 + z_2 + z_3 = 3z_0$

$(z_1 + z_2 + z_3)^2 = 9z_0^2$

$\Rightarrow z_1^2 + z_2^2 + z_3^2 + 2\left(z_1^2 + z_2^2 + z_3^2\right) = 9z_0^2$

$\Rightarrow z_1^2 + z_2^2 + z_3^2 = 3z_6^2$

$\sum_{k=1}^{3}\left(z_k - z_0\right)^2 = (z_1 - z_0)^2 + (z_2 - z_0)^2 + (z_3 - z_0)^2$

$= z_1^2 + z_2^2 + z_3^2 + 3z_0^2 - 2\left(z_1 + z_2 + z_3\right)z_0$

$= 6z_0^2 - 6z_0^2$

$= 0$

311. Let the product of $\omega_1 = (8 + i)\sin\theta + (7 + 4i)\cos\theta$ and $\omega_2 = (1 + 8i)\sin\theta + (4 + 7i)\cos\theta$ be $\alpha + i\beta, i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then p + q is equal to :

माना $\omega_1 = (8 + i)\sin\theta + (7 + 4i)\cos\theta$ और $\omega_2 = (1 + 8i)\sin\theta + (4 + 7i)\cos\theta$, $i = \sqrt{-1}$, का गुणनफल $\alpha + i\beta$ है। माना $\alpha + \beta$ के अधिकतम और न्यूनतम मान क्रमशः p और q हैं। तो p + q बराबर है :

(A) 160 (B) 150 (C) 140 (D) 130

Ans. D

Sol. $w_1w_2 = (8 + i)(1 + 8i)\sin^2\theta + (8 + i)(4 + 7i)\sin\theta\cos\theta + (7 + 4i)(1 + 8i)\sin\theta\cos\theta + (7 + 4i)(4 + 7i)\cos^2\theta$

$w_1w_2 = (65i)\sin^2\theta + (120i)\sin\theta\cos\theta + (65i)\cos^2\theta$

$= 65i + 60i(\sin2\theta)$

$\alpha = 0, \beta = 65 + 60\sin2\theta$

$p = 125, q = 5$

$p + q = 130$

312. Among the statements

(S1) : The set $\left\{z \in \mathbb{C} - \{-i\} : |z| = 1 \text{ and } \frac{z - i}{z + i} \text{ is purely real }\right\}$ contains exactly two elements, and

(S2) : The set $\left\{z \in \mathbb{C} - \{-1\} : |z| = 1 \text{ and } \frac{z - 1}{z + 1} \text{ is purely imaginary}\right\}$ contains infinitely many elements.

(A) both are incorrect (B) only (S2) is correct

(C) only (S1) is correct (D) both are correct

कथनों

(S1) : समुच्चय $\{z \in \mathbb{C} - \{-i\} : |z| = 1$ और $\frac{z-i}{z+i}$ शुद्धतः वास्तविक हैं$\}$ में तथ्यतः दो अवयव हैं, और

(S2) : समुच्चय $\{z \in \mathbb{C} - \{-1\} : |z| = 1$ और $\frac{z-1}{z+1}$ शुद्धतः काल्पनिक हैं$\}$ में अनन्त अवयव हैं

में से

(A) दोनों गलत हैं (B) केवल (S2) सही है

(C) केवल (S1) सही है (D) दोनों सही हैं

Ans. B

Sol. $S1 : \frac{z-i}{z+i} = \frac{\bar{z}+i}{\bar{z}-i}$

$\Rightarrow z + \bar{z} = 0$ So S1 is false

$S2 : \frac{z-1}{z+1} + \frac{\bar{z}-1}{\bar{z}+1} = 0$

$\Rightarrow \frac{z-1}{z+1} + \frac{\frac{1}{z}-1}{\frac{1}{z}+1} = 0 \Rightarrow \frac{z-1}{z+1} + \frac{1-z}{1+z} = 0$

$\Rightarrow 0 = 0$ So S2 is correct.

313. If the locus of $z \in C$, such that $\text{Re}\left(\frac{z-1}{2z+i}\right) + \text{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$, is a circle of radius r and center (a, b), then $\frac{15ab}{r^2}$ is equal to :

यदि $z \in C$, जिसके लिए $\text{Re}\left(\frac{z-1}{2z+i}\right) + \text{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$ है, का बिन्दुपथ r त्रिज्या और केन्द्र (a, b) का एक वृत्त है, तो $\frac{15ab}{r^2}$ बराबर है :

(A) 12 (B) 24 (C) 18 (D) 16

Ans. C

Sol. Let $\frac{z-1}{2z+i} = c$

$\text{Re}(c) + \text{Re}(\bar{c}) = 2\text{Re}(c)$

put $z = x + iy$

$$c=\frac{x+iy-1}{2x+2iy+i}=\frac{(x-1)+iy}{2x+i(2y+1)}$$

$$=\frac{((x-1)+iy)(2x-i(2y+1))}{4x^2+(2y+1)^2}$$

$$\Rightarrow \mathrm{Re}(c)=\frac{2x(x-1)+y(2y+1)}{4x^2+(2y+1)^2}$$

given $2\mathrm{Re}(c)=2$

$$\Rightarrow \frac{2x^2-2x+2y^2+y}{4x^2+4y^2+4y+1}=1$$

$$\Rightarrow x^2+y^2+x+\frac{3y}{2}+\frac{1}{2}=0$$

Centre $\left(-\frac{1}{2},-\frac{3}{4}\right)$ and $r^2=\frac{1}{4}+\frac{9}{16}-\frac{1}{2}=\frac{5}{16}$

$$\Rightarrow \frac{15ab}{r^2}=15\times\frac{1}{2}\times\frac{3}{4}\times\frac{16}{5}=18$$

314. Let α be a solution of $x^2+x+1=0$, and for some a and $b\in R$, $\begin{bmatrix}4 & a & b\end{bmatrix}\begin{bmatrix}1 & 16 & 13\\ -1 & -1 & 2\\ -2 & -14 & -8\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\end{bmatrix}$. If $\frac{4}{\alpha^4}+\frac{m}{\alpha^a}+\frac{n}{\alpha^b}=3$, then $m+n$ is equal to :

माना समीकरण $x^2+x+1=0$ का एक हल α है और किसी $a, b\in R$ के लिए, $\begin{bmatrix}4 & a & b\end{bmatrix}\begin{bmatrix}1 & 16 & 13\\ -1 & -1 & 2\\ -2 & -14 & -8\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\end{bmatrix}$ है। यदि $\frac{4}{\alpha^4}+\frac{m}{\alpha^a}+\frac{n}{\alpha^b}=3$, तो $m+n$ बराबर है :

(A) 8 (B) 7 (C) 3 (D) 11

Ans. D

Sol. $4-a-2b=0$, $64-a-14b=0$, $52+2a-8b=0$.

$\Rightarrow$ $a=-6, b=5$

Now,

$$\frac{4}{\alpha^4}+\frac{m}{\alpha^a}+\frac{n}{\alpha^b}=3 \qquad \Rightarrow \frac{4}{\omega^4}+\frac{m}{\omega^{-6}}+\frac{n}{\omega^5}=3$$

$4\omega^2 + m + n\omega = 3.$

$\Rightarrow \quad \left(\frac{m-3}{4}\right)+\left(\frac{n}{4}\right)\omega+\omega^2=0$

Clearly, $\frac{m-3}{4}=1, \quad \frac{n}{4}=1 \quad \left(\because 1+\omega+\omega^2=0\right)$

$m = 7, \ n = 4$

315. Let $A=\left\{\theta\in[0,2\pi]:1+10\,\mathrm{Re}\left(\frac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}\right)=0\right\}$. Then $\sum_{\theta\in A}\theta^2$ is equal to :

माना $A=\left\{\theta\in[0,2\pi]:1+10\,\mathrm{Re}\left(\frac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}\right)=0\right\}$ है। तो $\sum_{\theta\in A}\theta^2$ बराबर है :

(A) $6\pi^2$ (B) $8\pi^2$ (C) $\frac{27}{4}\pi^2$ (D) $\frac{21}{4}\pi^2$

Ans. D

Sol. Real part of $\frac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}=\frac{2\cos^2\theta-3\sin^2\theta}{\cos^2\theta+9\sin^2\theta}$

Now, $1+10\left(\frac{2\cos^2\theta-3\sin^2\theta}{\cos^2\theta+9\sin^2\theta}\right)=0$

$\cos^2\theta + 9\sin^2\theta + 20\cos^2\theta - 30\sin^2\theta = 0$

$\tan^2\theta = 1$

$\Rightarrow \quad \theta=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

$\sum_{\theta\in}\theta^2=\frac{\pi^2}{16}+\frac{9\pi^2}{16}+\frac{25\pi^2}{16}+\frac{49\pi^2}{16}=\frac{21\pi^2}{4}$

316. Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1)=\frac{-\pi}{4}$, $\arg(z_2)=0$ and $\arg(z_3)=\frac{\pi}{4}$.

If $\left|z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1\right|^2=\alpha+\beta\sqrt{2},\alpha,\beta\in Z$, then the value of $\alpha^2 + \beta^2$ is :

माना कि z_1, z_2 और z_3, $\arg(z_1)=\frac{-\pi}{4}$, $\arg(z_2)=0$ और $\arg(z_3)=\frac{\pi}{4}$ के साथ वृत्त $|z| = 1$ पर तीन सम्मिश्र संख्याएँ हैं। यदि $\left|z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1\right|^2=\alpha+\beta\sqrt{2},\alpha,\beta\in Z$ है, तो $\alpha^2 + \beta^2$ का मान है :

(A) 31 (B) 24 (C) 29 (D) 41

Ans. C

Sol. $Z_1 = e^{-i\frac{\pi}{4}}, Z_2 = 1, Z_3 = e^{i\frac{\pi}{4}}$

$$\left|z_1\overline{z}_2 + z_2\overline{z}_3 + z_3\overline{z}_1\right|^2 = \left|e^{-i\frac{\pi}{4}} \times 1 + 1 \times e^{-i\frac{\pi}{4}} + e^{i\frac{\pi}{4}} \times e^{i\frac{\pi}{4}}\right|^2$$

$$\left|e^{-i\frac{\pi}{4}} + e^{-i\frac{\pi}{4}} + e^{i\frac{\pi}{4}}\right|^2$$

$$= \left|2e^{-\frac{\pi}{4}} + i\right|^2 = \left|\sqrt{2} - \sqrt{2}i + i\right|^2$$

$$= \left(\sqrt{2}\right)^2 + \left(1 - \sqrt{2}\right)^2 = 2 + 1 + 2 - 2\sqrt{2} = 5 - 2\sqrt{2}$$

$\alpha = 5, \beta = 2$

$\Rightarrow \alpha^2 + \beta^2 = 29$

317. Let the curve $z(1 + i) + \overline{z}(1 - i) = 4$, $z \in C$, divide the region $|z - 3| \le 1$ into two parts of areas α and β. Then $|\alpha - \beta|$ is equal to :

माना कि वक्र $z(1 + i) + \overline{z}(1 - i) = 4$, $z \in C$, क्षेत्र $|z - 3| \le 1$ को क्षेत्रफलों α और β के दो भागों में विभाजित करता है। तब $|\alpha - \beta|$ बराबर है :

(A) $1 + \frac{\pi}{2}$ (B) $1 + \frac{\pi}{6}$ (C) $1 + \frac{\pi}{3}$ (D) $1 + \frac{\pi}{4}$

Ans. A

Sol. $z(1 + i) + \overline{z}(1 - i) = 4$

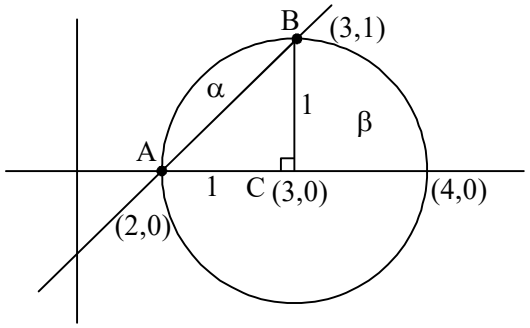
put $x = x + iy$

$(x + iy)(1 + i) + (x - iy)(1 - i) = 4$

$x - y + iy + ix + x - ix - iy - y = 4$

$x - y = 2$

$|Z - 3| \le 1$ circle



$\alpha + \beta = \pi r^2 = \pi(1)^2$

$\alpha + \beta = \pi$

Now α = Area of sector – Area of ΔABC

$$= \frac{\pi(1)^2}{4} - \frac{1}{2} \times 1 \times 1$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

So, $\beta = \pi - \alpha$

$|\alpha - \beta| = |2\alpha - \pi|$

$$= \left|\frac{\pi}{2} - 1 - \pi\right|$$

$$= \left|-1 - \frac{\pi}{2}\right| = 1 + \frac{\pi}{2}$$

318. Let $\left|\frac{\bar{z} - i}{2\bar{z} + i}\right| = \frac{1}{3}$, $z \in C$, be the equation of a circle with center at C. If the area of the triangle,whose vertices are at the points (0,0), C and (α, 0) is 11 square units, then α^2 equals :

माना कि $\left|\frac{\bar{z} - i}{2\bar{z} + i}\right| = \frac{1}{3}$, $z \in C$, केन्द्र C के साथ एक वृत्त का समीकरण है। यदि उस त्रिभुज, जिसके शीर्ष, बिनदुओं (0,0), C और (α, 0) पर हैं, का क्षेत्रफल 11 वर्ग इकाई है, तो α^2 बराबर है :

(A) 50 (B) $\frac{121}{25}$ (C) 100 (D) $\frac{81}{25}$

Ans. C

Sol. $3|x - iy - i| = |2(x - iy) + i|$

$3|x - i(y + 1)| = |2x + i(1 - 2y)|$

$$3\sqrt{x^2 + (y+1)^2} = \sqrt{4x^2 + (1-2y)^2}$$

$9x^2 + 9y^2 + 18y + 9 = 4x^2 + 4y^2 - 4y + 1$

$5x^2 + 5y^2 + 22y + 8 = 0$

$$x^2 + y^2 + \frac{22}{5}y + \frac{8}{5} = 0$$

$$C\left(0, -\frac{11}{5}\right)$$

$$\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 0 & -\frac{11}{5} & 1 \\ \alpha & 0 & 1 \end{vmatrix} = 11$$

$\left|\frac{11\alpha}{5}\right| = 22$

$\left|\frac{\alpha}{5}\right| = 2$

$\alpha^2 = 100$

319. The number of complex numbers z, satisfying $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$, is :

$|z| = 1$ और $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ को संतुष्ट करने वाली सम्मिश्र संख्याओं z की संख्या है :

(A) 6 (B) 4 (C) 10 (D) 8

Ans. D

Sol. $|z| = 1$ & $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$

Let $z = e^{i\theta}$

$\bar{z} = e^{-i\theta}$ $\quad\therefore \frac{z}{\bar{z}} = \frac{e^{i\theta}}{e^{-i\theta}} = e^{2i\theta}$

$|e^{2i\theta} + e^{-i2\theta}| = 1$

$|\cos 2\theta + i \sin 2\theta + \cos 2\theta - i \sin 2\theta| = 1$

$|\cos 2\theta| = \frac{1}{2}$

$\cos 2\theta = \pm\frac{1}{2}$

Number of solution $\equiv 8$

320. If α and β are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then

$$16 \cdot \mathrm{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \cdot \mathrm{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \text{ is equal to :}$$

यदि α और β, समीकरण $2z^2 - 3z - 2i = 0$, जहाँ $i = \sqrt{-1}$ है, के मूल हैं, तो

$$16 \cdot \mathrm{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \cdot \mathrm{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \text{ बराबर है :}$$

(A) 398 (B) 409 (C) 441 (D) 312

Ans. C

Sol. α is a root of $2z^2 - 3z - 2i = 0$

$\therefore \quad 2\alpha^2 - 3\alpha - 2i = 0$

$\Rightarrow \quad 2\alpha^2 - 2i = 3\alpha$

$\Rightarrow \alpha - \frac{i}{\alpha} = \frac{3}{2}$

$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} - 2i = \frac{9}{4}$

$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} = \frac{9}{4} + 2i$

$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} - 2 = \frac{81}{16} - 4 + 9i$

$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} = \frac{81-32}{16} + 9i = \frac{49}{16} + 9i$

$\Rightarrow$ Similarly $\beta^4 + \frac{1}{\beta^4} = \frac{49}{16} + 9i$

Now, $\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} = \frac{\alpha^{15}\left(\alpha^4 + \frac{1}{\alpha^4}\right) + \beta^{15}\left(\beta^4 + \frac{1}{\beta^4}\right)}{\alpha^{15} + \beta^{15}}$

$= \frac{\left(\frac{49}{16} + 9i\right)\left(\alpha^{15} + \beta^{15}\right)}{\alpha^{15} + \beta^{15}} = \frac{49}{16} + 9i$

$\therefore \quad$ Ans. $= 16 \times \frac{49}{16} \times 9 = 441$

321. Let O be the origin, the point A be $z_1 = \sqrt{3} + 2\sqrt{2}i$, the point B($z_2$) be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then :

(A) ABO is an obtuse angled isosceles triangle (B) ABO is a scalene triangle

(C) area of triangle ABO is $\frac{11}{4}$ (D) area of triangle ABO is $\frac{11}{\sqrt{3}}$

माना कि O मूल बिन्दु, बिन्दु A, $z_1 = \sqrt{3} + 2\sqrt{2}i$, बिन्दु B($z_2$) इस प्रकार हैं कि $\sqrt{3}|z_2| = |z_1|$ और $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$ है। तब :

(A) ABO एक अधिककोण समद्विबाहु त्रिभुज है। (B) ABO एक विषमबाहु त्रिभुज है।

(C) त्रिभुज ABO का क्षेत्रफल $\frac{11}{4}$ है। (D) त्रिभुज ABO का क्षेत्रफल $\frac{11}{\sqrt{3}}$ है।

Ans. A

Sol. $z_1 = \sqrt{3} + 2\sqrt{2}i$ & $\frac{|z_2|}{|z_1|} = \frac{1}{\sqrt{3}}$

given $\arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{6}$

$$z_2 = \frac{|z_2|}{|z_1|} \cdot z_1 e^{i\left(\frac{\pi}{6}\right)}$$

$$z_2 = \frac{1}{\sqrt{3}} \cdot \frac{\left(\sqrt{3} + 2\sqrt{2}i\right)\left(\sqrt{3} + i\right)}{2}$$

$$z_2 = \frac{\left(3 - 2\sqrt{2}\right) + i\left(2\sqrt{6} - \sqrt{3}\right)}{2\sqrt{3}}$$

Now,

$$z_1 - z_2 = \frac{\left(3 + 2\sqrt{2}\right) + i\left(2\sqrt{6} - \sqrt{3}\right)}{2\sqrt{3}}$$

$|z_1 - z_2| = |z_2| \Rightarrow \Delta ABO$ is isosceles with angles $\frac{\pi}{6}, \frac{\pi}{6}$ & $\frac{2\pi}{3}$

322. Let $|z_1 - 8 - 2i| \le 1$ and $|z_2 - 2 + 6i| \le 2, z_1, z_2 \in C$. Then minimum value of $|z_1 - z_2|$ is :

माना कि $|z_1 - 8 - 2i| \le 1$ तथा $|z_2 - 2 + 6i| \le 2, z_1, z_2 \in C$ है। तब $|z_1 - z_2|$ का निम्नतम मान है :

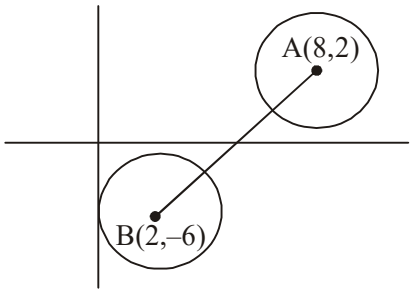
(A) 7 (B) 3 (C) 10 (D) 13

Ans. A

Sol. $|z_1 - (8 + 2i)| \le 1$ $|z_2 - (2 - 6i)| \le 2$

$C_1 : (8, 2)$ $C_2 : (2, -6)$

$r_1 = 1$ $r_2 = 2$



AB = 10

$|z_1 - z_2|_{min} = 10 - 2 - 1 = 7$

323. Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0$, $z \in C$. Then $4(\alpha^2 + \beta^2)$ is equal to :

माना समीकरण $(\bar{z})^2 + |z| = 0$, $z \in C$ के सभी शून्येत्तर हलों के योग और गुणनफल क्रमशः α और β हैं। तो $4(\alpha^2 + \beta^2)$ बराबर है:

(A) 2 (B) 8 (C) 4 (D) 6

Ans. C

Sol. $z = x + iy$

$\bar{z} = x - iy$

$\bar{z}^2 = x^2 - y^2 - 2ixy$

$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x = 0$ or $y = 0$

$-y^2 + |y| = 0$ $x^2 + |x| = 0$

$|y| = |y|^2$ $x = 0$

$y = 0, \pm 1$

$\Rightarrow i, -i$ $\Rightarrow \alpha = i - i = 0$

are roots $\beta = i(-i) = 1$

$4(0 + 1) = 4$

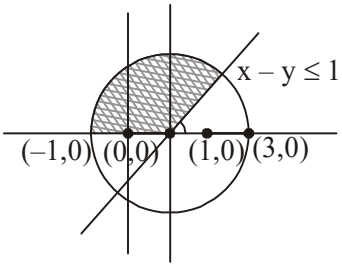
324. The area (in sq. units) of the region $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, Im(z) \ge 0\}$ is :

क्षेत्र $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, Im(z) \ge 0\}$ का क्षेत्रफल (वर्ग इकाईयों में) है :

(A) $\frac{7\pi}{4}$ (B) $\frac{7\pi}{3}$ (C) $\frac{17\pi}{8}$ (D) $\frac{3\pi}{2}$

Ans. D

Sol.



$|z-1| \le 2 \Rightarrow (x-1)^2 + y^2 = 4$

$(z+\bar{z})+i(z-\bar{z}) \le 2 \Rightarrow x - y \le 1$

$\text{Im}\,|z| \ge 0 \Rightarrow y \ge 0$

$$\text{Area of region} = \frac{1}{2}(2)^2\left(\frac{3\pi}{4}\right) = \frac{3\pi}{2}$$

325. Consider the following two statements

Statement I : For any two non-zero complex numbers z_1, z_2.

$$(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2(|z_1|+|z_2|) \text{ and}$$

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$, then $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y} = 1$.

Between the above two statemetns :

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

कथनों

कथन I : किसी भी दो शून्येत्तर सम्मिश्र संख्याओं z_1, z_2 के लिए

$$(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2(|z_1|+|z_2|) \text{ तथा}$$

कथन II : यदि x, y, z तीन भिन्न सम्मिश्र संख्याएँ हैं और a, b, c तीन धनात्मक वास्तविक संख्याएँ हैं ताकि $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

है, तो $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y}=1$ है।

में से :

(A) कथन I सही है, परन्तु कथन II गलत है

(B) कथन I गलत है परन्तु कथन II सही हैं

(C) कथन I तथा कथन II दोनों गलत है

(D) कथन I तथा कथन II दोनों सही है

Ans. A

Sol. **(I)** $(|Z_1|+|Z_2|)\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right|$

$$\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right|\le\left|\frac{Z_1}{|Z_1|}\right|+\left|\frac{Z_2}{|Z_2|}\right|=1+1=2$$

(II) $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

$$\frac{a^2}{(y-z)}\times\left(\frac{\overline{y-z}}{y-z}\right)+\frac{b^2}{z-x}\times\frac{\overline{z-x}}{z-x}+\frac{c^2}{x-y}\times\frac{\overline{x-y}}{x-y}$$

$$=\frac{a^2}{|y-z|^2}(\bar{y}-\bar{z}+\bar{z}-\bar{x}+\bar{x}-\bar{y})=0$$

So not true

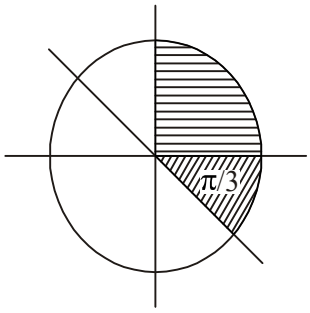
326. Let $S_1=\{z\in C:|z|\le 5\}, S_2\left\{z\in C:\text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ and $S_3=\{z\in C:\text{Re}(z)\ge 0\}$. Then the area of the region $S_1\cap S_2\cap S_3$ is :

माना $S_1=\{z\in C:|z|\le 5\}, S_2\left\{z\in C:\text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ तथा $S_3=\{z\in C:\text{Re}(z)\ge 0\}$ हैं। तो क्षेत्र $S_1\cap S_2\cap S_3$ का क्षेत्रफल है :

(A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{4}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{6}$

Ans. C

Sol.



$S_1 : x^2 + y^2 \le 5^2$; $S_2 : \sqrt{3}\ x + y \ge 0$; $S_3 : x \ge 0$

$$\frac{z+1-i\sqrt{3}}{1-\sqrt{3}i} = \frac{\left((x+1)+i\left(y-\sqrt{3}\right)\left(1+i\sqrt{3}\right)\right)}{1+3}$$

$$I_m(z) = (x+1)\sqrt{3} + \left(y-\sqrt{3}\right) = \sqrt{3}\,x + y$$

$$S_1 \cap S_2 \cap S_3 = \frac{1}{4}\pi 5^2 + \frac{1}{2}5^2\frac{\pi}{3} = \frac{125\pi}{12}$$

327. If z_1, z_2 are two distinct complex number such that $\left|\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2}\right| = 2$, then :

(A) both z_1 and z_2 lie on the same circle

(B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1

(C) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

(D) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1

यदि z_1, z_2 दो भिन्न सम्मिश्र संख्याऐं इस प्रकार है कि $\left|\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2}\right| = 2$ है, तो :

(A) z_1 तथा z_2 दोनों समान वृत्त पर स्थित हैं

(B) z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर तथा z_2, 1 त्रिज्या के वृत्त पर स्थित है

(C) या तो z_1, 1 त्रिज्या के वृत्त पर या z_2, $\frac{1}{2}$ त्रिज्या के वृत्त पर स्थित है

(D) या तो z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर या z_2, 1 त्रिज्या के वृत्त पर स्थित है

Ans. C

Sol. $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\overline{z}_2}\right|=2$

$\Rightarrow |z_1-2z_2|=|1-2z_1\overline{z}_2|$

$\Rightarrow |z_1-2z_2|^2|1-2z_1\overline{z}_2|^2$

$\Rightarrow |z_2|^2-2z_1\overline{z}_2-2\overline{z}_1z_2+4|z_2|^2=1-2\overline{z}_1z_2-2z_1\overline{z}_2+4|z_1|^2|z_2|^2$

$\Rightarrow |z_1|^2-1+4|z_2|^2-4|z_1|^2|z_2|^2=0$

$\Rightarrow \left(|z_1|^2-1\right)\left(1-4|z_2|^2\right)=0$

either $|z_1|^2-1=0$ or $1-4|z_2|^2=0$

$|z_1|=1$ or $|z_2|=\frac{1}{2}$

$\therefore$ either z_1 lies on circle of radius 1

or z_2 lies on circle of radius $\frac{1}{2}$

328. Let z be a complex number such that $|z+2|=1$ and $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $\left|\text{Re}\left(\overline{z+2}\right)\right|$ is :

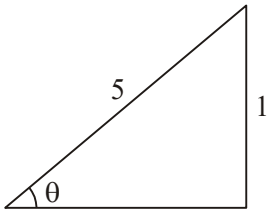
माना z एक समिश्र संख्या इस प्रकार है कि $|z+2|=1$ और $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$ तब $\left|\text{Re}\left(\overline{z+2}\right)\right|$ का मान होगा :

(A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{1+\sqrt{6}}{5}$ (D) $\frac{\sqrt{6}}{5}$

Ans. A

Sol. Let $z+2=e^{i\theta} \Rightarrow \left(\frac{z+1}{z+2}\right)=\frac{-2+e^{i\theta}+1}{e^{i\theta}}=1-\frac{1}{e^{i\theta}}$

$\Rightarrow \left(\frac{z+1}{z+2}\right)=1-\cos\theta+i\sin\theta$



$\sin\theta=\frac{1}{5}$

$\text{Re}(z+2)=\cos\theta \Rightarrow \left|\text{Re}\left(\overline{z+2}\right)\right|=\frac{2\sqrt{6}}{5}$

329. If the set $R=\{(a, b): a+5b=42, a, b \in N\}$ has m elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is :

यदि समुच्चय $R = \{(a, b) : a + 5b = 42, a, b \in N\}$ में m अवयव है और $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, जहाँ $i=\sqrt{-1}$, तब $m + x + y$ का मान होगा :

(A) 8 (B) 12 (C) 5 (D) 4

Ans. B

Sol. $a + 5b = 42, a, b \in N$

$a = 42 - 5b, b = 1, a = 37$

$b = 2, a = 32$

$b = 3, a = 27$

$\vdots$

$b = 8, a = 2$

R has "8" elements $\Rightarrow m = 8$

$\sum_{n=1}^{8}\left(1-i^{n!}\right)=x+iy$

for $n \geq 4$, $i^{n!} = 1$

$\Rightarrow (1 - i) + (1 - i^{2!}) + (1 - i^{3!})$

$= 1 - i + 2 + 1 + 1$

$= 5 - i = x + iy$

$m + x + y = 8 + 5 - 1 = 12$

330. Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z-(6+8i)|$ is equal to :

माना z एक सम्मिश्र संख्या है, जिसके लिए $\frac{z-2i}{z+2i}$ का वास्तविक भाग शून्य है। तो $|z-(6+8i)|$ का अधिकतम मान बराबर है :

(A) ∞ (B) 8 (C) 10 (D) 12

Ans. D

Sol. $\frac{z-2i}{z+2i}+\frac{\overline{z}+2i}{\overline{z}-2i}=0$

$z\overline{z}-2i\overline{z}-2iz+4(-1)+z\overline{z}+2zi+2\overline{z}i+4(-1)=0$

$\Rightarrow 2|z|^2=8$

$\Rightarrow |z|=2$

$|z-(6+8i)|_{maximum}=10+2=12$

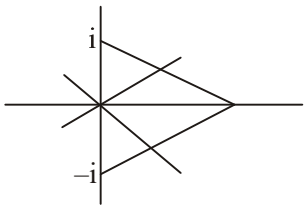
331. If $S=\{z\in C:|z-i|=|z+i|=|z-1|\}$, then n(S) is :

यदि If $S=\{z\in C:|z-i|=|z+i|=|z-1|\}$ है, तो n(S) बराबर है :

(A) 2 (B) 0 (C) 3 (D) 1

Ans. D

Sol.



332. If $z=x+iy, xy\neq 0$, satisfies the equation $z^2+i\overline{z}=0$, then $|z^2|$ is equal to :

यदि $z=x+iy, xy\neq 0$ समीकरण $z^2+i\overline{z}=0$ को संतुष्ट करता है, तो $|z^2|$ बराबर है :

(A) 1 (B) 1/4 (C) 9 (D) 4

Ans. A

Sol. $z=x+iy, x\neq 0, y\neq 0$ $\Rightarrow |z|\neq 0$

$z^2+i\overline{z}=0$

$\Rightarrow z^2 = -i\bar{z}$

$\Rightarrow |z^2| = |-i\bar{z}| = |-i||\bar{z}|$

$\Rightarrow |z|^2 = |z| \Rightarrow |z|(|z|-1) = 0$

$\Rightarrow |z| = 1$

333. If z is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to :

यदि z एक सम्मिश्र संख्या हैं, तो समीकरणों $z^{1985} + z^{100} + 1 = 0$ तथा $z^3 + 2z^2 + 2z + 1 = 0$ के उभयनिष्ठों मूलों की संख्या है :

(A) 0 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Given $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$

The roots of $z^{1985} + z^{100} + 1 = 0$ be ω and ω^2 and also satisfies $z^3 + 2z^2 + 2z + 1 = 0$

$\therefore$ ω and ω^2 are common solutions

(where ω is cube root of unity)

$\therefore$ 2 solutions

334. Let z_1 and z_2 be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals :

माना दो साम्मिश्र संख्याओं z_1 तथा z_2 के लिए $z_1 + z_2 = 5$ तथा $z_1^3 + z_2^3 = 20 + 15i$ हैं। तो $|z_1^4 + z_2^4|$ बराबर है :

(A) $30\sqrt{3}$ (B) $25\sqrt{3}$ (C) $15\sqrt{15}$ (D) 75

Ans. D

Sol. $z_1 + z_2 = 5$

$z_1^3 + z_2^3 = 20 + 15i$

$z_1^3 + z_2^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2)$

$z_1^3 + z_2^3 = 125 - 3z_1.z_2(5)$

$\Rightarrow 20 + 15i = 125 - 15z_1z_2$

$\Rightarrow 3z_1z_2 = 25 - 4 - 3i$

$\Rightarrow 3z_1z_2 = 21 - 3i$

$\Rightarrow z_1 \cdot z_2 = 7 - i$

$\Rightarrow (z_1 + z_2)^2 = 25$

$\Rightarrow z_1^2 + z_2^2 = 25 - 2(7 - i)$

$\Rightarrow 11 + 2i$

$\left(z_1^2+z_2^2\right)^2=121-4+44i$

$\Rightarrow z_1^4+z_2^4+2(7-i)^2=117+44i$

$\Rightarrow z_1^4+z_2^4=117+44i-2(49-1-14i)$

$\Rightarrow \left|z_1^4+z_2^4\right|=75$

335. Let $S=\{z\in C:|z-1|=1$ and $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}\}$. Let $z_1, z_2\in S$ be such that $|z_1|=\max\limits_{Z\in S}|z|$ and $|z_2|=\min\limits_{Z\in S}|z|$. Then $\left|\sqrt{2}z_1-z_2\right|^2$ equals :

माना $S=\{z\in C:|z-1|=1$ तथा $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}\}$ है। माना $z_1, z_2\in S$ के लिए $|z_1|=\max\limits_{Z\in S}|z|$ तथा $|z_2|=\min\limits_{Z\in S}|z|$ हैं। तो $\left|\sqrt{2}z_1-z_2\right|^2$ बराबर है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)(2x)-i(2iy)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)x+y=\sqrt{2}$

and circle $|z-1|=1$

$(x-1)^2+y^2=1$

$(x-1)^2+\left(\sqrt{2}-\left(\sqrt{2}-1\right)x\right)^2=1$

$(x-1)^2+\left(\sqrt{2}(1-x)+x\right)^2=1$

$(x-1)^2+2(1-x)^2+x^2+2x\sqrt{2}(1-x)=1$

$3(x-1)^2+x^2+2\sqrt{2}x(1-x)=1$

$\left(-2\sqrt{2}+4\right)x^2+x\left(-6+2\sqrt{2}\right)+2=0$

$x=1$ or $x=\dfrac{1}{2-\sqrt{2}}$

for $x=1\Rightarrow y=1$

and $x=\dfrac{1}{2-\sqrt{2}}\Rightarrow y=\sqrt{2}-\dfrac{1}{\sqrt{2}}$

So, $\left|\sqrt{2}z_1 - z_2\right|^2 = \left|\frac{1}{\sqrt{2}-1} + i - 1 - i\right|^2 = 2$

336. If z is a complex number such that $|z| \geq 1$, then the minimum value of $\left|z + \frac{1}{2}(3+4i)\right|$ is :

यदि एक सम्मिश्र संख्या z के लिए $|z| \geq 1$ है, तो $\left|z + \frac{1}{2}(3+4i)\right|$ का न्यूनतम मान है :

(A) 3 (B) $\frac{5}{2}$ (C) 2 (D) $\frac{3}{2}$

Ans. Bonus

Sol. p(z) is a point lying outside circle $x^2 + y^2 = 1$

$Q\left(\frac{-3}{2}, -2\right)$

$\left|z + \frac{1}{2}(3+4i)\right| = PQ$

$(PQ)_{min} = 0$ when P coincides with Q.

337. Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in \mathbb{C} : Re\left(az^2 + bz\right) = a\right.$ and $\left.Re\left(bz^2 + az\right) = b\right\}$ is equal to :

माना $a \neq b$ दो शून्येतर वास्तविक संख्याएँ है। तो समुच्चय $X = \left\{z \in \mathbb{C} : Re\left(az^2 + bz\right) = a\right.$ तथा $\left.Re\left(bz^2 + az\right) = b\right\}$ में अवयवों की संख्या है :

(A) 1 (B) 3 (C) 2 (D) 0

Ans. D

Sol. $Re\left(az^2 + bz\right) = a$

$az^2 + bz + a\bar{z}^2 + b\bar{z} = 2a$

$a\left(z^2 + \bar{z}^2\right) + b(z + \bar{z}) = 2a$ .....(1)

$Re\left(bz^2 + az\right) = b$

$bz^2 + az + b\bar{z}^2 + a\bar{z} = 2b$

$b\left(z^2 + \bar{z}^2\right) + a(z + \bar{z}) = 2b$ .....(2)

$(1) \times b - (2) \times (a)$

$\Rightarrow \left(b^2 - a^2\right)(z + \bar{z}) = 0$

$\Rightarrow \quad (z + \bar{z}) = 0 \quad \left(a^2 \neq b^2\right)$

$(1) \times a - (2) \times (b)$

$\Rightarrow \quad \left(a^2 - b^2\right)(z + \bar{z}) = 2\left(a^2 - b^2\right) \quad \left(a^2 \neq b^2\right)$

$z^2 + \bar{z}^2 = 2$

$\Rightarrow (z + \bar{z})^2 - 2z\bar{z} = 2$

$z\bar{z} = -1$

$\Rightarrow 1 + 1^2 = -1$

$\Rightarrow$ No solution

But when $a = -b$,

$\text{Re}(az - az) = a$

$\Rightarrow \quad \text{Re}\left(a\left(x^2 - y^2 + i2xy\right) - a(x + iy)\right) = a$

$\Rightarrow a\left(x^2 - y^2\right) - ax = a$

$\Rightarrow x^2 - y^2 - x = 1$

$\Rightarrow x^2 - x - 1 = y^2$

For any real values of y there two values of x, hence infinite complex numbers are possible.

338. Let the complex number $z = x + iy$ be such that $\frac{2z - 3i}{2z + i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to :

माना सम्मिश्र संख्या $z = x + iy$ के लिए $\frac{2z - 3i}{2z + i}$ मात्र काल्पनिक है। यदि $x + y^2 = 0$ है, तो $y^4 + y^2 - y$ बराबर है :

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Ans. C

Sol. $z = x + iy$

$\frac{(2z - 3i)}{2z + i} = \text{purely imaginary}$

means $\text{Re}\left(\frac{2z - 3i}{2z + i}\right) = 0$

$\Rightarrow \frac{(2z - 3i)}{(2z + i)} = \frac{2(x + iy) - 3i}{2(x + iy) + i}$

$$=\frac{2x+2yi-3i}{2x+i2y+i}$$

$$=\frac{2x+i(2y-3)}{2x+i(2y+1)}\times\frac{2x-i(2y+1)}{2x-i(2y+1)}$$

$$\text{Re}\left[\frac{2z-3i}{2z+i}\right]=\frac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2}=0$$

$$\Rightarrow 4x^2+(2y-3)(2y+1)=0$$

$$\Rightarrow 4x^2+\left[4y^2+2y-6y-3\right]=0$$

$$\because x+y^2=0\Rightarrow x=-y^2$$

$$\Rightarrow 4\left(-y^2\right)^2+4y^2-4y-3=0$$

$$\Rightarrow 4y^4+4y^2-4y=3$$

$$\Rightarrow y^4+y^2-y=\frac{3}{4}$$

Therefore, correct answer is option (3).

339. Let $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i}\text{ is a real number}\right\}$. Then which of the following is NOT correct :

माना $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i}\text{ एक वास्तविक संख्या}\right\}$ । तो निम्न में कौन सा सही नहीं है ?

(A) $y+x^2+y^2\neq-\frac{1}{4}$ (B) $x=0$

(C) $(x,y)=\left(0,-\frac{1}{2}\right)$ (D) $y\in\left(-\infty,-\frac{1}{2}\right)\cup\left(-\frac{1}{2},\infty\right)$

Ans. C

Sol. $\because\quad z=x+iy$ and $\frac{2z-3i}{4z+2i}$ is real number

$\therefore\quad \frac{2x+(2y-3)i}{4x+(4y+2)i}$ is real number.

$\therefore\quad \frac{(2x+(2y-3)i)(4x-(4y+2)i)}{(4x)^2+(4y+2)^2}$ is real number

$\therefore\quad -2x(4y+2)+(2y-3)4x=0$

$\therefore\quad x=0$

Here $x = 0$ and $y \in \mathbb{R}$ but $y \neq -\frac{1}{2}$ is not acceptable

in this case

Denominator will be zero.

340. Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of $w_1 - w_2$ is equal to :

माना $z_1 = 5 + 4i$ को मूल बिंदु के सापेक्ष घड़ी की विपरीत दिशा में एक समकोण तक घुमाने पर बिंदु w_1 प्राप्त होता है तथा $z_2 = 3 + 5i$ को मूलबिंदु के सापेक्ष घड़ी की दिशा में एक समकोण तक घुमाने पर बिंदु w_2 प्राप्त होता है। तो $w_1 - w_2$ का मुख्य आयाम बराबर है :

(A) $-\pi + \tan^{-1}\frac{8}{9}$ (B) $\pi - \tan^{-1}\frac{33}{5}$ (C) $-\pi + \tan^{-1}\frac{33}{5}$ (D) $\pi - \tan^{-1}\frac{8}{9}$

Ans. D

Sol. $W_1 = z_i i = (5+4i)i = -4+5i \ldots$ (i)

$W_1 = z_2(-i) = (3+5i)(-i) = 5-3i \ldots (2)$

$W_1 - W_2 = -9+8i$

Principal argument $= \pi - \tan^{-1}\left(\frac{8}{9}\right)$

341. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : Re(a+\bar{z}) > Im(\bar{a}+z)\}$ and $B = \{z \in \mathbb{C} : Re(a+\bar{z}) < Im(\bar{a}+z)\}$. Then among the two statements :

(S1) : If Re (a), Im (a) > 0, then the set A contains all the real numbers

(S2) : If Re (a), Im (a) < 0, then the set B contains all the real numbers

(A) only (S1) is true (B) both are false (C) only (S2) is true (D) both are true

$a \in \mathbb{C}$ के लिए, माना $A = \{z \in \mathbb{C} : Re(a+\bar{z}) > Im(\bar{a}+z)\}$ तथा $B = \{z \in \mathbb{C} : Re(a+\bar{z}) < Im(\bar{a}+z)\}$ हैं। तो दो कथनों :

(S1) : यदि Re (a), Im (a) > 0 है, तो सभी वास्तविक संख्याएँ A में हैं

(S2) : यदि Re (a), Im (a) < 0 है, तो सभी वास्तविक संख्याएँ B में हैं

(A) केवल (S1) सही है (B) दोनो गलत है (C) केवल (S2) सही है (D) दोनो सही है

Ans. B

Sol. Let $a = x_1 + iy_1, z = x + iy$

Now $Re(a+\bar{z}) > Im(\bar{a}+z)$

$\therefore x_1 + x > -y_1 + y$

$x_1 = 2, y_1 = 10, x = -12, y = 0$

Given inequality is not valid for these values.

S1 is false.

Now $\text{Re}(a+\bar{z}) < \text{Im}(\bar{a}+z)$

$x_1 + x < -y_1 + y$

$x_1 = -2, y_1 = -10, x = 12, y = 0$

Given inequality is not valid for these values.

S2 is false.

342. Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1+3i)$ and radius r = 1. Let $z_1 = 1 + i$ and the complex number z_2 be outside the circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to :

माना सम्मिश्र तल में वृत C का केन्द्र $z_0 = \frac{1}{2}(1+3i)$ तथा त्रिज्या r = 1 हैं। माना $z_1 = 1 + i$ है तथा वृत्त के बाहर सम्मिश्र संख्या z_2 इस प्रकार है कि $|z_1 - z_0||z_2 - z_0| = 1$ है। यदि z_0, z_1 तथा z_2 संरेख हैं, तो $|z_2|^2$ का छोटा मान बराबर है :

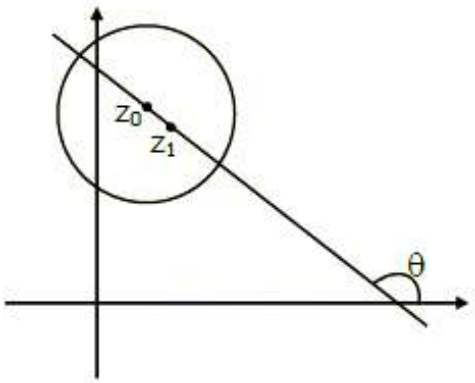
(A) $\frac{7}{2}$ (B) $\frac{13}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. D

Sol. $|z_1 - z_0| = \left|\frac{1-i}{2}\right| = \frac{1}{\sqrt{2}}$

$\Rightarrow |z_2 - z_\circ| = \sqrt{2};$ centre $\left(\frac{1}{2}, \frac{3}{2}\right)$

$z_\circ\left(\frac{1}{2}, \frac{3}{2}\right)$ and $z_1(1,1)$



$\tan\theta = -1 \Rightarrow \theta = 135^\circ$

$z_2\left(\frac{1}{2} + \sqrt{2}\cos 135^\circ, \frac{3}{2} + \sqrt{2}\sin 135^\circ\right)$

or

$\left(\frac{1}{2} - \sqrt{2}\cos 135^\circ, \frac{3}{2} - \sqrt{2}\sin 135^\circ\right)$

$\Rightarrow z_2\left(-\frac{1}{2}, \frac{5}{2}\right)$ or $z_2\left(\frac{3}{2}, \frac{1}{2}\right)$

$\Rightarrow |z_2|^2 = \frac{26}{4}, \frac{5}{2}$

$\Rightarrow |z_2|_{\min^2} = \frac{5}{2}$

343. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to :

माना समीकरण $x^2 - \sqrt{2}x + 2 = 0$ के मूल α, β हैं। तो $\alpha^{14} + \beta^{14}$ बराबर है :

(A) -64 (B) -128 (C) $-64\sqrt{2}$ (D) $-128\sqrt{2}$

Ans. B

Sol. $x^2 - \sqrt{2}x + 2 = 0$

$$x = \frac{\sqrt{2} \pm \sqrt{2-8}}{2} = \frac{\sqrt{2} \pm \sqrt{6}i}{2}$$

$$\alpha = \frac{\sqrt{2} + \sqrt{6}i}{2} = \sqrt{2}e^{\frac{i\pi}{3}} \ \& \ \beta = \sqrt{2}e^{\frac{-i\pi}{3}}$$

$$\alpha^{14} = 2^7 e^{\frac{i14\pi}{3}} = 128\left[e^{\frac{i2\pi}{3}}\right]$$

$$\beta^{14} = 128\left[e^{\frac{-i2\pi}{3}}\right]$$

$$\alpha^{14} + \beta^{14} = 128(2)\cos\left(\frac{2\pi}{3}\right) = -128$$

344. Let $S = \left\{z \in \mathbb{C} : \bar{z} = i\left(z^2 + \text{Re}(\bar{z})\right)\right\}$. The $\sum_{z \in S} |z|^2$ is equal to :

माना $S = \left\{z \in \mathbb{C} : \bar{z} = i\left(z^2 + \text{Re}(\bar{z})\right)\right\}$ है। तो $\sum_{z \in S} |z|^2$ बराबर है :

(A) $\frac{5}{2}$ (B) 4 (C) 3 (D) $\frac{7}{2}$

Ans. B

Sol. Let $z = x + iy$

$$\bar{z} = i\left(z^2 + \text{Re}\left(\bar{z}\right)\right)$$

$\Rightarrow x - iy = i(x^2 - y^2 + 2ixy + x)$

$x - iy = i(x^2 - y^2 + x) - 2xy$

$x = -2xy \Rightarrow x(2y + 1) = 0$

$$\Rightarrow x = 0, y = \frac{-1}{2} \quad \text{....(i)}$$

$-y = x^2 - y^2 + x$ ....(ii)

Case (1) $x = 0$

(ii) $\Rightarrow -y = -y^2 \Rightarrow y^2 - y = 0 \Rightarrow y = 0, 1$

$z = 0, i$

Case (2) $y = \frac{-1}{2}$

(ii) $\Rightarrow \frac{1}{2} = x^2 - \frac{1}{4} + x \Rightarrow x^2 + x - \frac{3}{4} = 0$

$4x + 4x - 3 = 0 \Rightarrow (2x - 1)(2x + 3) = 0$

$x = \frac{1}{2}, \frac{-3}{2}$

$z = \frac{1}{2} - \frac{1}{2}i, \frac{-3}{2} - \frac{1}{2}i$

$\sum |z|^2 = 0 + 1 + \frac{1}{2} + \frac{5}{2} = 4$

345. $A = \{z \in C : |z - 2 - i| = 3\}, B = \{z \in C : Re(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to_____.

माना $A = \{z \in C : |z - 2 - i| = 3\}, B = \{z \in C : Re(z - iz) = 2\}$ और $S = A \cap B$ हैं। तो $\sum_{z \in S} |z|^2$ बराबर है _____.

Ans. 22

Sol. $A = \{z \in C : |z - 2 - i| = 3\}$

$B = \{z \in C : Re(z - iz) = 2\}$

$S = A \cap B$

then $\sum_{z \in S} |z|^2 = ??$

Sol. $A : (x - 2)^2 + (y - 1)^2 = 9$ ...(i)

$B : Re\{(x + y) + i(y - x)\} = 2$

$x + y = 2$ ...(ii)

Solving (i) & (ii)

$x = \frac{3 \pm \sqrt{17}}{2}; \; y = \frac{1 \mp \sqrt{17}}{2}$

$\sum_{z \in S} |z|^2 = \frac{1}{4}[2 \times 26 + 2 \times 18]$

$\Rightarrow \frac{88}{4}$

$\Rightarrow 22$

346. If α is a root of the equation $x^2 + x + 1 = 0$ and $\sum_{k=1}^{n}\left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$, then n is equal to_____.

यदि समीकरण $x^2 + x + 1 = 0$ का एक मूल α है और $\sum_{k=1}^{n}\left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$ है, तो n बराबर है_____.

Ans. 11

Sol. $\alpha = w, w^2$

$$\sum_{k=1}^{n}\left(w^k + \frac{1}{w^k}\right)^2 = 20$$

$$\left(w + \frac{1}{w}\right)^2 + \left(w^2 + \frac{1}{w^2}\right)^2 + \left(w^3 + \frac{1}{w^3}\right)^2 + \left(w^4 + \frac{1}{w^4}\right)^2 + ... = 20$$

$\Rightarrow (w + w^2)^2 + (w^2 + w)^2 + 2^2 + (w + w^2)^2 + ... = 20$

$\Rightarrow (-1)^2 + (-1)^2 + 4 + (-1)^2 + ... = 20$

$\Rightarrow n = 11$

347. If $\alpha = 1 + \sum_{r=1}^{6}(-3)^{r-1}\ {}^{12}C_{2r-1}$, then the distance of the point $\left(12, \sqrt{3}\right)$ from the line $\alpha x - \sqrt{3}y + 1 = 0$ is _____.

यदि $\alpha = 1 + \sum_{r=1}^{6}(-3)^{r-1}\ {}^{12}C_{2r-1}$ है, तो बिन्दु $\left(12, \sqrt{3}\right)$ की रेखा $\alpha x - \sqrt{3}y + 1 = 0$ से दूरी है _____.

Ans. 5

Sol. $\alpha = 1 + \sum_{r=1}^{6}(-1)^{r-1}\ {}^{12}C_{2r-1}3^{r-1}$

$$\alpha = 1 + \sum_{r=1}^{6}{}^{12}C_{2r-1}\frac{\left(\sqrt{3}i\right)^{2t-1}}{\sqrt{3}i} \qquad i = \text{iota, let } \sqrt{3}i = x$$

$$\alpha = 1 + \frac{1}{\sqrt{3}i}\left({}^{12}C_1 x + {}^{12}C_3 x^3 + ...{}^{12}C_{11}x^{11}\right)$$

$$= 1 + \frac{1}{\sqrt{3}i}\left(\frac{\left(1 + \sqrt{3}i\right)^{12} - \left(1 - \sqrt{3}i\right)^{12}}{2}\right)$$

$$= 1 + \frac{1}{\sqrt{3}i}\left(\frac{\left(-2w^2\right)^{12} - \left(2w\right)^{12}}{2}\right) = 1$$

So, distance of $\left(12,\sqrt{3}\right)$ from $x-\sqrt{3}y+1=0$ is

$$\frac{12-3+1}{2}=5$$

348. Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right|=1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=1, z\in C$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to _____.

माना कि पूर्णांक $a, b \in [-3, 3]$ इस प्रकार है कि $a + b \neq 0$ है। तब उन सभी संभव क्रमित युग्मों (a, b), जिनके लिए $\left|\frac{z-a}{z+b}\right|=1$ और $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=1, z\in C$ हैं , जहाँ ω और ω^2 समीकरण $x^2 + x + 1 = 0$ के मूल हैं, की संख्या बराबर बराबर है_____.

Ans. 10

Sol. $a, b \in I, -3 \le a, b \le 3, a + b \neq 0$

$|z - a| = |z + b|$

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=1$$

$$\Rightarrow \begin{vmatrix} z & z & z \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=1$$

$$\Rightarrow z\begin{vmatrix} 1 & 1 & 1 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=1$$

$$\Rightarrow z\begin{vmatrix} 1 & 0 & 0 \\ \omega & z+\omega^2-\omega & 1-\omega \\ \omega^2 & 1-\omega^2 & z+\omega-\omega^2 \end{vmatrix}=1$$

$\Rightarrow z^3 = 1$

$\Rightarrow z = \omega, \omega^2, 1$

Now $|1 - a| = |1 + b|$

$\Rightarrow$ 10 pairs

349. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right) = \frac{125}{16}m$. Then the value of m is _____.

माना x_1, x_2, x_3, x_4 समीकरण $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ के हल हैं तथा $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right) = \frac{125}{16}m$ है। तो m का मान है _____

Ans. 221

Sol. $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$

x_1, x_2, x_3, x_4 are the solution of equation

$4x^4 + 8x^3 - 17x^2 - 12x + 9 = 4(x - x_1)(x - x_2)(x - x_3)(x - x_4)$

put $x = 2i$

$4(2i)^4 + 8(2i)^3 - 17(2i)^2 - 12(2i) + 9$

$= 4[(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)]$

$\Rightarrow \quad 64 - 64i + 68 - 24i + 9$

$\Rightarrow \quad |141 - 88i| = 4|(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)|$

$\Rightarrow \quad (141)^2 + (88)^2 = 16[(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2)]$

$\Rightarrow \quad \frac{19881 + 7744}{16} = \left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_4^2\right)$

$\Rightarrow \quad \frac{27625}{16} = \frac{125}{16}m$

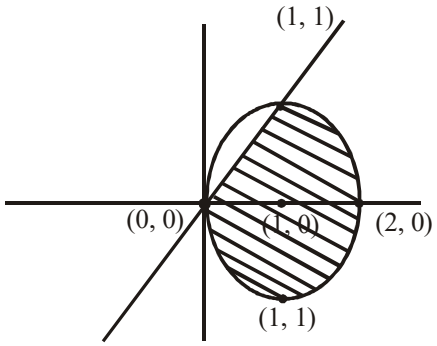
$m = 221$

350. The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ is _____.

समुच्चय $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ के अवयवों के मापांक के वर्गों का योग है _____.

Ans. 9

Sol.



$|Z-1| \le 1$

$|Z-5| \le |Z-5i|$

(a, b) = {(0, 0), (1, 1), (1, 0), (1, –1), (2, 0)}

$|Z|^2$ = {0, 2, 1, 2, 4|

Sum = 9

351. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \ge 0$, then $5(3A - 2B - C)$ is equal to _____.

यदि α समीकरण $x^2 + x + 1 = 0$ को संतुष्ट करता है तथा $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \ge 0$ हैं, तो $5(3A - 2B - C)$ बराबर है _____।

Ans. 5

Sol. $\alpha^2 + \alpha + 1 = 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1 \left(x^2 + x + 1 = 0\right) 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1$ let $\alpha = \omega$

$(1+\alpha)^7 = 1 + {}^7c_1\alpha + {}^7c_2\alpha^2 + {}^7c_3\alpha^3 + {}^7c_4\alpha^4 + {}^7c_5\alpha^5 + {}^7c_6\alpha^6 + {}^7c_7\alpha^7$

$= 1 + 7\alpha + 21\alpha^2 + 35\alpha^3 + 35\alpha^4 + 24\alpha^5 + 7\alpha^6 + \alpha^9$

$\because \alpha = \omega$

$= 1 + 7\omega + 21\omega^2 + 35\omega^3 + 35\omega^4 + 21\omega^5 + 7\omega^6 + \omega^7$

$\left(\because \omega^3 = 1\right)$

$= 1 + 7\omega + 21\omega^2 + 35 + 35\omega + 21\omega + 7 + \omega$

$= 43 + 43\omega + 42\omega^2$

$= 43 + 43\alpha + 42\alpha^2 = A + B\alpha + C\alpha^2$

$\Rightarrow$ A = 43 |B = 43| C = 42

5(129 – 86 – 42)

=5(129 – 128) = 5

352. Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circle $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where

$z_0 = 1 + i$. Then, the value of $100\,|\alpha|^2$ is _____.

माना सम्मिश्र संख्याएँ α तथा $\frac{1}{\bar{\alpha}}$ क्रमशः वृत्तों $|z - z_0|^2 = 4$ तथा $|z - z_0|^2 = 16$ पर हैं, जहाँ $z_0 = 1 + i$ है। तो $100\,|\alpha|^2$ का मान है _____।

Ans. 20

Sol. $|\alpha - z_0|^2 = 4$

$\Rightarrow (\alpha - z_0)(\bar{\alpha} - \bar{z}_0) = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |z_0|^2 = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 = 2$ _____(i)

(ii) $\left|\frac{1}{\bar{\alpha}} - z_0\right|^2 = 16 \Rightarrow (1 - \bar{\alpha}z_0)(1 - \alpha\bar{z}_0) = 16\,|\alpha|^2$

$\Rightarrow 1 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |\alpha|^2 \cdot 2 = 16\,|\alpha|^2$ _____(ii)

from (i) and (ii) $-1 - |\alpha|^2 = 2 - 16\,|\alpha|^2 \Rightarrow 15\,|\alpha|^2 = 3 \Rightarrow 5\,|\alpha|^2 = 1$

$100|\alpha|^2 = 20$

353. Let α, β the roots of the equation $x^2 - x + 2 = 0$ with $\text{Im}(\alpha) > \text{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

माना समीकरण $x^2 - x + 2 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। तो $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ बराबर है_____.

Ans. 13

Sol. $x^2 + 2 = x$

$\Rightarrow x^4 + 4 + 4x^2 = x^2$

So,

$x^4 = -4 - 3x^2$

$\Rightarrow x^6 = -4x^2 - 3x^4$

$\Rightarrow x^6 = 12 + 5x^2$

Now,

$\alpha^6 - 5\alpha^2 = 12$

So $\alpha^4 + \beta^4 + 12$

$= 4 - 3(\alpha^2 + \beta^2)$

$= 4 - 3\,[1 - 4]$

$= 13$

354. Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\text{Im}(\alpha) > \text{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$. Then n + a + b is equal to _____.

माना समीकरण $x^2 - \sqrt{6}x + 3 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। माना 3 से अभाज्य पूर्णांक a, b हैं तथा घन पूर्णांक n के लिए $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$ है। तो n + a + b बराबर है _____

Ans. 49

Sol. $x^2 - \sqrt{6}x + 3 = 0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$

$$x = \frac{\sqrt{6} \pm i\sqrt{6}}{2} = \frac{\sqrt{6}}{2}(1 \pm i)$$

$$\alpha = \sqrt{3}\left(e^{i\frac{\pi}{4}}\right),\ \beta = \sqrt{3}\left(e^{-i\frac{\pi}{4}}\right)$$

$$\therefore \frac{\alpha^{99}}{\beta} + \alpha^{98} = \alpha^{98}\left(\frac{\alpha}{\beta} + 1\right)$$

$$= \frac{\alpha^{98}(\alpha + \beta)}{\beta} = 3^{49}\left(e^{i99\frac{\pi}{4}}\right) \times \sqrt{2}$$

$= 3^{49}(-1 + i)$

$= 3^n(a + ib)$

$\therefore n = 49, a = -1, b = 1$

$\therefore n + a + b = 49 - 1 + 1 = 49$

355. If α denotes the number of solutions of $|1 - i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)}\right)$, where $z = \frac{\pi}{4}(1 + i)^4\left[\frac{1 - \sqrt{\pi}i}{\sqrt{\pi} + i} + \frac{\sqrt{\pi} - i}{1 + \sqrt{\pi}i}\right]$, $i = \sqrt{-1}$, then the distance of the point (α, β) from the line $4x - 3y = 7$ is _____.

यदि $|1 - i|^x = 2^x$ के हलों की संख्या α है तथा $\beta = \left(\frac{|z|}{\arg(z)}\right)$ है, जहाँ $z = \frac{\pi}{4}(1+i)^4\left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right]$, $i = \sqrt{-1}$ है, तो रेखा $4x - 3y = 7$ से बिन्दु (α, β) की दूरी है _____.

Ans. 3

Sol. $z = \frac{\pi}{4}(1+i)^4\left[\frac{1+\pi+\pi+1}{i(1+\pi)-\sqrt{\pi}+\sqrt{\pi}}\right]$

$= \frac{\pi}{4}(1+i)^4\frac{2}{i} = \frac{\pi}{2}\frac{(1+i)^4}{i} = \frac{-\pi}{2}i(1+i)^4$

$|z| = \left|\frac{-\pi}{2}\right||i||1+i|^4 = 2\pi$

$z = \frac{-\pi}{2}(-4i) = 2\pi i \Rightarrow \arg(z) = \frac{\pi}{2}$

$\Rightarrow \beta = \frac{2\pi}{\frac{\pi}{2}} = 4$

Also $|1 - i|^x = 2^x$

$\left(\sqrt{2}\right)^x = 2^x$

$\Rightarrow x = 0$

$\Rightarrow$ 1 solution $\Rightarrow \alpha = 1$

Now perpendicular distance of $4x - 3y = 7$ from (1,4)

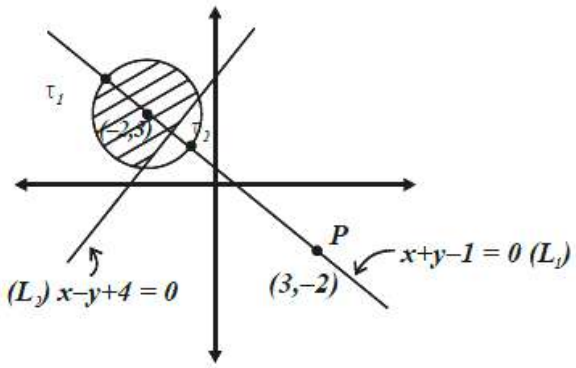
$\left|\frac{4(1)-3(4)-7}{5}\right| = 3$

356. Let $P = \{z \in C : |z+2-3i| \le 1\}$ and $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$. Let in $P \cap Q$, $|z-3+2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where α, β are integers, then $\alpha + \beta$ equals _____.

माना $P = \{z \in C : |z + 2 - 3i| \le 1\}$ तथा $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$ हैं। माना $P \cap Q$ में, $|z - 3 + 2i|$, z_1 तथा z_2 पर क्रमशः अधिकतम तथा निम्नतम है। यदि $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है _____

Ans. 36

Sol.



Clearly for the shaded region z_1 is the intersection of the circle and the line passing through P (L_1) and z_2 is intersection of line L_1 & L_2

Circle : $(x + 2)^2 + (y - 3)^2 = 1$

$L_1 : x + y - 1 = 0$

$L_2 : x - y + 4 = 0$

On solving circle & L_1 we get

$$z_1 : \left(-2 - \frac{1}{\sqrt{2}}, 3 + \frac{1}{\sqrt{2}}\right)$$

On solving L_1 and z_2 is intersection of line L_1 & L_2

we get $z_2 : \left(\frac{-3}{2}, \frac{5}{2}\right)$

$$|z_1|^2 + 2|z_2|^2 = 14 + 5\sqrt{2} + 17$$

$$= 31 + 5\sqrt{2}$$

So, $\alpha = 31$

$\beta = 5$

$\alpha + \beta = 36$